

Christelle Seri

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EXPERIENCE

ERCOT | Application Developer II

Austin, Texas

July 2023 - Present

- Primary developer for a **\$2.5M** power grid communications project owning end-to-end **Java backend architecture** and database design
- Led backend performance optimization by tuning database queries and implementing pagination and indexing, reducing CPU utilization from **100% to 40%** during load testing
- Designed and implemented **20+ SQL database tables** with constraints, procedures, and indexing strategies
- Established the automated test suite, building **45 unit and integration tests with JUnit** covering messaging success and error scenarios
- Designed and implemented **31 REST API endpoints** supporting core messaging workflows, documented using **OpenAPI**
- Rebuilt the core functionality of the legacy Servlet/JDBC application as a **Spring Boot** and **Hibernate** proof of concept, implementing request validation, pagination, **LDAP authentication**, CSV export, and global exception handling
- Worked directly with ERCOT operators to translate operational requirements and user feedback into application designs

PROJECTS

Event-Driven Notification System

[GitHub](#)

- Built a distributed notification system using two **Spring Boot microservices** communicating asynchronously through **RabbitMQ** with a transactional outbox pattern for reliable event publication
- Improved messaging reliability through **retry with exponential backoff, dead-letter queues**
- Implemented end-to-end integration tests using **Mockito, MockMvc, and Testcontainers** to validate asynchronous messaging workflows
- Containerized the microservices with **Docker Compose** and deployed the application to an **AWS EC2** instance

EDUCATION

University of Texas at Austin

August 2024 - May 2026

Masters of Science in Data Science

GPA: 3.73

University of Illinois at Urbana-Champaign

August 2019 - May 2023

Bachelor of Science in Electrical Engineering with Honors

GPA: 3.78

SKILLS

- **Programming Languages:** Java, SQL, Python, JavaScript
- **Databases:** Oracle, PostgreSQL
- **Frameworks:** Spring Boot, Spring Security, Hibernate, JUnit, Mockito, MockMvc, Testcontainers, React
- **Tools & Platforms:** Docker, RabbitMQ, Flyway, Git, Apache Tomcat, AWS (EC2), Maven